

Executive Summary for Government Stakeholders

The project “Customer Segmentation & Churn Pattern Analytics in European Banking” was conducted to analyze customer churn behavior and identify important factors influencing customer retention in the banking sector. Customer churn refers to customers discontinuing banking services, which can negatively impact profitability, customer relationships, and financial stability.

The banking sector generates large amounts of customer-related data, but understanding churn behavior across different customer groups remains a challenge. Most institutions focus on overall churn percentages without detailed segmentation analysis. As a result, retention strategies often remain general rather than targeted.

The purpose of this study was to understand how customer churn varies across different customer segments based on demographics, geography, financial profile, and engagement levels. The analysis focused on customers from European banking regions including France, Germany, and Spain.

The project used customer-level banking data containing variables such as age, geography, credit score, tenure, account balance, estimated salary, customer activity, and churn status. Exploratory Data Analysis (EDA) techniques were applied to identify trends and hidden churn patterns among customer groups.

The study found that customer churn behavior differed significantly across customer segments. Geography-wise analysis showed differences in churn among European countries, indicating that customer retention challenges may vary by region. Customers with low credit scores showed slightly higher churn compared to customers with medium or high credit scores.

Similarly, inactive customers demonstrated higher churn behavior compared to active customers, suggesting that customer engagement plays an important role in customer retention. The analysis also revealed that high-balance customers experienced relatively higher churn, which may create financial risk for banks because premium customers contribute significantly to profitability.

Customer tenure analysis indicated that new customers showed slightly higher churn, highlighting the importance of strong onboarding experiences and early relationship management. These findings suggest that banks should adopt customer-segment-specific strategies instead of relying on generalized retention methods.

To improve understanding and accessibility of the analysis, a Streamlit web application dashboard was developed. The dashboard included modules such as:

- Overall churn summary

- Geography-wise churn visualization
- Age and tenure churn comparison
- High-value customer churn explorer

The dashboard also provided interactive features such as segment filters, dynamic KPI updates, and drill-down analysis, helping users understand churn patterns more effectively.

Based on the findings, several recommendations were proposed. Banks should improve customer engagement among inactive customers, strengthen retention efforts for premium customers, improve onboarding for new customers, and develop geography-specific retention strategies.

Overall, this project demonstrates how customer segmentation and data analytics can help better understand customer churn in banking. The findings of the study may support decision-makers in designing more effective customer retention strategies and improving long-term banking relationships.